



Data Infrastructure Compliance

Practical 12-Point Checklist for Robot Operators and Fleet Managers

Including Audit Trails, Integrity
Protection, Data Minimisation and
Selective Replication

How to solve **Annex I Data Requirements** for ROS Fleets using
Specialized Time-Series Storage.

ReductStore – High Performance
Time-Series Blob Storage for
Robotics & Industrial AI

Executive Summary

The EU Cyber Resilience Act (CRA) is coming fast:

Vulnerability reporting obligation



STARTS FROM 11 SEP 2026

Full compliance

(Security by Design, 5-year lifecycle support, CE marking)



MANDATORY FROM 11 DEC 2027

ROS-based robotics fleets are classified as “products with digital elements” and often fall into Class II. Operators and manufacturers must prove auditability, data integrity, access control and data minimisation for all stored and transmitted sensor data.

Most current solutions (rosvbag2, MinIO, generic TSDBs) force you to choose between **compliance** and **performance**.

This checklist gives you a clear, practical framework to evaluate your current setup and shows how a specialised time-series blob storage like ReductStore can make your ROS fleets’ Data Infrastructure CRA-ready — while delivering 10× faster ingestion, instant time-range replay and up to 90 % lower cloud costs.

Data Infrastructure CRA Requirements Relevant for ROS Fleets (Overview)

CRA ANNEX I REQUIREMENT	RELEVANCE FOR ROS FLEETS	COMMON PAIN POINT TODAY
 Protection from unauthorised access	Access to raw sensor data and logs	Weak or missing granular logging
 Integrity protection & corruption reporting	Sensor data and ROS bags must remain unaltered	No reliable proof of tampering
 Data minimisation	Only necessary data may be stored/replicated	Everything is sent to the cloud
 Logging & monitoring	Who accessed which data and when	No record-level audit trails
 Security updates & lifecycle support	Minimum 5 years support	Complex versioning of storage layer

The 12-Point CRA Checklist for ROS Fleets (1-2)

1

Token-based Authentication & Granular Access Control

REQUIREMENT	Every device, service or user must have scoped access rights.
WHY IT MATTERS	Prevents unauthorised access to sensitive sensor data.
REDUCTSTORE SOLUTION	Native token-based authentication with granular permissions at Bucket, Entry and Record level.

Self-check: Do we have per-user / per-device access control with logging?

☐ Yes ☐ No

[→ Comments](#)

2

Per-Record Audit Logging

REQUIREMENT	Full traceability of who read, wrote or replicated which record.
WHY IT MATTERS	Core CRA requirement for incident investigation and conformity assessment.
REDUCTSTORE SOLUTION	Automatic audit trail on every Read/Write/Replication action (timestamp + token + label + action).

Self-check: Do we have record-level audit logs?

☐ Yes ☐ No

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Checklist for ROS Fleets

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3

Data Integrity Protection & Corruption Detection

REQUIREMENT	Proof that stored ROS data has not been tampered with.
WHY IT MATTERS	Critical for safety-relevant systems.
REDUCTSTORE SOLUTION	Label-based cryptographic integrity checks + automatic corruption reporting per block.

Self-check: Can we prove data integrity for every stored record?

☐ Yes ☐ No

[→ Comments](#)

4

Label-Based Data Minimisation & Selective Replication

REQUIREMENT	Only necessary data may be stored and transferred.
WHY IT MATTERS	Reduces attack surface and cloud costs.
REDUCTSTORE SOLUTION	Conditional, label-based replication (e.g. only “high-risk” events + context window).

Self-check: Do we replicate only relevant data?

☐ Yes ☐ No

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Checklist for ROS Fleets

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5

Encryption at Rest and in Transit

REQUIREMENT	Protection of confidentiality.
WHY IT MATTERS	Sensor data often contains sensitive information.
REDUCTSTORE SOLUTION	Client-side and server-side encryption options with seamless SROS2 integration.

Self-check: Is all data encrypted at rest and in transit?

☐ Yes ☐ No

[→ Comments](#)

6

Version Traceability & Secure Update Mechanisms

REQUIREMENT	Lifecycle support and secure updates for at least 5 years.
WHY IT MATTERS	Mandatory for long-term compliance.
REDUCTSTORE SOLUTION	Versioned replication and dedicated security update paths.

Self-check: Can we securely update the storage layer?

☐ Yes ☐ No

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Checklist for ROS Fleets

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7

SBOM Generation

REQUIREMENT Software Bill of Materials for the storage component.

WHY IT MATTERS Required for conformity assessment.

REDUCTSTORE SOLUTION Automated SBOM export for ReductStore itself.

Self-check: Do we have an SBOM for our data infrastructure?

☐ Yes ☐ No

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8

Vulnerability Handling & Reporting Readiness

REQUIREMENT Processes for handling and reporting actively exploited vulnerabilities.

WHY IT MATTERS 24h / 72h reporting obligation from Sep 2026.

REDUCTSTORE SOLUTION Dedicated security channels and audit-ready logging.

Self-check: Are we prepared for CRA vulnerability reporting?

☐ Yes ☐ No

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Checklist for ROS Fleets

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9

Edge-First Architecture with FIFO Quotas

REQUIREMENT	Resilient operation under real industrial constraints.
WHY IT MATTERS	Robots have limited storage and connectivity.
REDUCTSTORE SOLUTION	FIFO quotas and intelligent batching to prevent disk overruns..

Self-check: Do we protect edge devices from storage overflow?

☐ Yes ☐ No

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10

Fast Time-Range Replay for Incident Investigation

REQUIREMENT	Quick access to historical data for root-cause analysis.
WHY IT MATTERS	Essential for debugging and compliance.
REDUCTSTORE SOLUTION	Native time-series indexing for exact time-range queries across multimodal data.

Self-check: Can we replay exact moments within seconds?

☐ Yes ☐ No

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Checklist for ROS Fleets

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11 Seamless ROS 2 Integration

REQUIREMENT	Native support for robotics data formats.
WHY IT MATTERS	Most fleets run ROS 2.
REDUCTSTORE SOLUTION	Dedicated ROS 2 Agent + SROS2 compatibility.

Self-check: Is our storage natively integrated with ROS 2?

☐ Yes ☐ No

[→ Comments](#)

12 Audit-Ready Export & Documentation Generation

REQUIREMENT	Ability to generate technical documentation for conformity assessment.
WHY IT MATTERS	CE marking and audits.
REDUCTSTORE SOLUTION	Automated export of audit logs, SBOMs and compliance reports.

Self-check: Can we generate audit-ready documentation easily?

☐ Yes ☐ No

[→ Comments](#)

Quick-Start Migration Guide

(3 Steps)

01

Deploy ReductStore Agent on your robots
(Docker/binary, < 5 min).



02

Configure label-based replication rules and FIFO quotas.



03

Connect existing tools (Grafana, Foxglove) and run first benchmark.



Next Steps & ROI

Typical results from production fleets using ReductStore

up to
90%

lower cloud
costs

10×
15×

10× faster ingestion /
15× faster reads

**Audit
Ready**

Full Data
Infrastructure CRA
audit readiness

NEXT STEP

Ready to make your ROS fleet data pipeline CRA-compliant?

Book your **free 20-minute Production Fleet Audit** today. We will analyse your current data pipeline and run a live benchmark with your own ROS data.

[Book a Production Fleet Audit →](#)

<https://www.reduct.store/demo>

